

DISNEY+

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Team 8

OVERVIEW

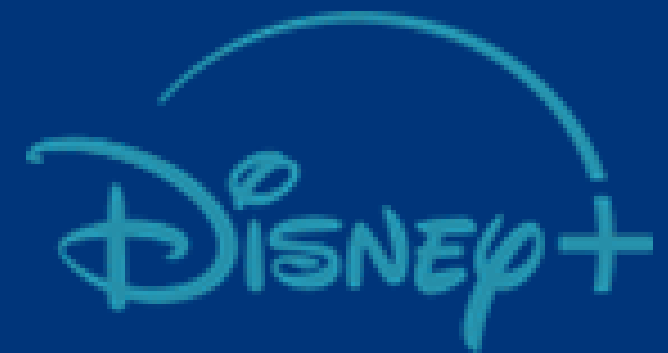
Disney+ is a subscription video platform for films, series, and originals from Disney brands.

Households subscribe to a subscription plan which offer ad-free or ad-supported plans, a maximum number of register devices, and a maximum number profiles.

The system is personalized via watch activity and enforces parental control.

Implementation of a database schema for management of various entities such as:

- Accounts
- Subscription plans
- Content
- Advertisers
- Playback events



Structure the data efficiently and ensure smooth relationships between tables while supporting scalability for the growing amount of data typical in streaming platforms.

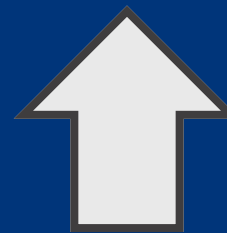
BUSINESS REQUIREMENTS

The content catalog is represented by a supertype content (identified by **ContentID**) with common attributes **ReleaseYear**, **Rating**, and **Language**.

There are two subtypes: **Movie** and **Episode**.

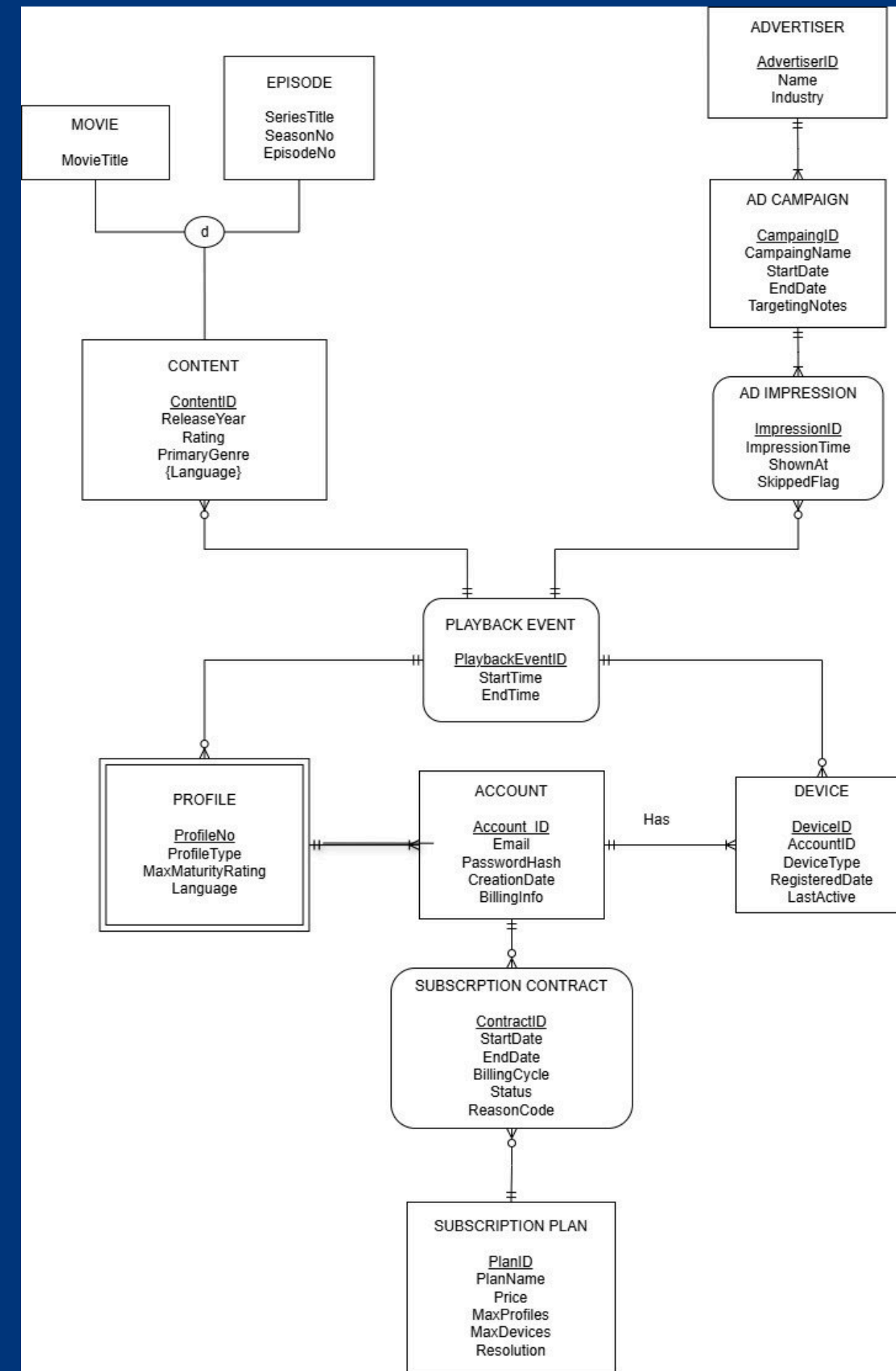
Movie has a **MovieTitle**. **Episode** has episodes with **EpisodeNo**, **SeasonNo** and **SeriesTitle**.

We also dropped the subtypes of the **AdSubscriptionPlan**

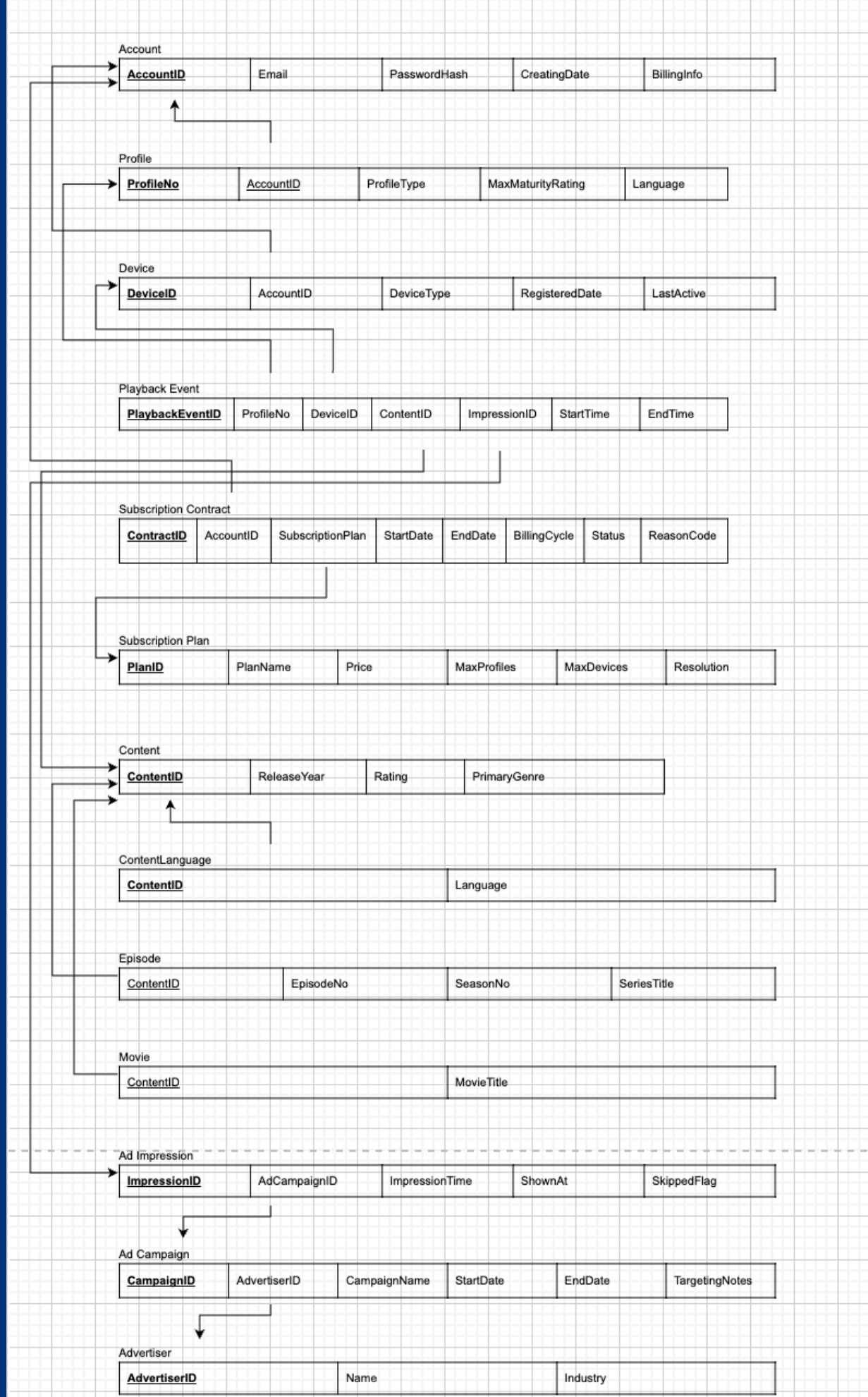


The content catalog is represented by a supertype content (identified by **ContentID**) with common attributes **TitleName**, **PrimaryGenre**, **Languages**, **ReleaseYear**, **Rating**, and **IsOriginal (Y/N)**. There are two subtypes: **Movie** and **Series**. A **Movie** has a **MovieDuration**; a **Movie** is directly playable. A **Series** has episodes with **SeasonNo**, **EpisodeNo**, **EpisodeDuration** and **EpisodeReleaseDate** and belongs to exactly one **Series**.

ER DIAGRAM



LD DIAGRAM



QUERY FUNCTION 1

Total watch minutes and play events per movie/series episode title within a time period

Given the title of the movie or series episode and its content type, list the total minutes watched (computed from start time to end time) and the number of playback events during October 2025.

This function returns the **Title**, **Content Type**, **Total Minutes Watched**, and **Number of Playback Events** for all content played between 2025-10-01 00:00:00 and 2025-11-01 00:00:00.

The content team could track the popularity of titles during different time periods to inform future planning and promotional strategies.

```
SELECT
  T.Title,
  T.ContentType,
  SUM(TIMESTAMPDIFF(MINUTE,
    PE.start_time,
    PE.end_time)) AS TotalMinutesWatched,
  COUNT(*) AS PlayEvents
FROM playback_event PE
  INNER JOIN
    (SELECT
      m.content_id, m.movie_title AS Title, 'Movie' AS ContentType
    FROM movie m UNION ALL SELECT
      e.content_id,
      e.series_title AS Title,
      'Episode' AS ContentType
    FROM
      episode e) AS T ON T.content_id = PE.content_id
WHERE PE.start_time >= '2025-10-01 00:00:00'
  AND PE.start_time < '2025-11-01 00:00:00'
GROUP BY T.Title, T.ContentType
ORDER BY TotalMinutesWatched DESC;
```

	Title	ContentType	TotalMinutesWatched	PlayEvents
▶	Ahsoka	Episode	1260	24
	The Mandalorian	Episode	1197	24
	Soul	Movie	957	10
	Togo	Movie	921	9
	Hawkeye	Episode	894	18

QUERY FUNCTION 2

Per-title ad performance for a given campaign

Given an Ad Campaign ID, list each title that served ads under that campaign and report the total impressions, skipped impressions, skip rate (%), and unique profiles reached.

This function returns the **Title**, **Impressions**, **Skipped**, **Skip Rate (%)**, and **Unique Profiles** for all ad impressions tied to the specified campaign.

The advertising operations team and advertiser managers could assess the delivery and engagement of each title within a campaign.

```
SELECT
  AI.campaign_id,
  T.Title,
  COUNT(*) AS Impressions,
  SUM(CASE WHEN AI.skipped_flag = 1 THEN 1 ELSE 0 END) AS Skipped,
  ROUND(100 * SUM(CASE WHEN AI.skipped_flag = 1 THEN 1 ELSE 0 END) / COUNT(*), 1) AS SkipRatePct,
  COUNT(DISTINCT PE.account_id, PE.profile_no) AS UniqueProfiles
FROM ad_impression AI
JOIN playback_event PE
  ON AI.impression_id = PE.impression_id
JOIN (
  SELECT m.content_id, m.movie_title AS Title FROM movie m
  UNION ALL
  SELECT e.content_id, e.series_title AS Title FROM episode e
) AS T
  ON T.content_id = PE.content_id
GROUP BY AI.campaign_id, T.Title;
```

	campaign_id	Title	Impressions	Skipped	SkipRatePct	UniqueProfiles
▶	7001	Soul	1	0	0.0	1
	7002	Andor	3	3	100.0	1
	7002	Hawkeye	3	0	0.0	1
	7002	The Mandalorian	3	0	0.0	1
	7003	Andor	3	3	100.0	1
	7003	Hawkeye	3	0	0.0	1
	7003	Loki	3	0	0.0	1
	7004	Ahsoka	6	3	50.0	2

QUERY FUNCTION 3

Device Type Usage for a given period of time

Given all devices registered in the system, list each device type and report its usage activity during a given time period (October 2025).

This function returns the **Device Type**, **Total Playback Events**, **Total Unique Accounts**, and **Total Unique Profiles** for all playback activity that occurred between 2025-10-01 00:00:00 and 2025-11-01 00:00:00.

The technical operations team could analyze the most frequently used platforms and allocate development and support resources more efficiently.

```
SELECT
    D.device_type AS DeviceType,
    COUNT(PE.playback_event_id) AS PlayEvents,
    COUNT(DISTINCT PE.account_id) AS UniqueAccounts,
    COUNT(DISTINCT PE.profile_no) AS UniqueProfiles
FROM
    device D
    JOIN playback_event PE ON D.device_id = PE.device_id
    AND D.account_id = PE.account_id
WHERE
    PE.start_time >= '2025-10-01 00:00:00'
    AND PE.start_time < '2025-11-01 00:00:00'
GROUP BY D.device_type
ORDER BY PlayEvents DESC;
```

	DeviceType	PlayEvents	UniqueAccounts	UniqueProfiles
▶	Laptop	17	9	3
	Smart TV	16	9	3
	Android Phone	9	5	3
	iPad	7	4	3

QUERY FUNCTION 4

Language Availability Count for Movies and Episodes

Given movies and episodes in the database, list each title with its content type and the total number of languages available for that content.

This function returns the Title, Content Type (Movie or Episode), and Language Count for all content items by counting the languages in the content_language table.

The translation team could more effectively track language availability for each title and prioritize translation efforts accordingly.

```
SELECT
  T.Title,
  T.ContentType,
  COUNT(DISTINCT CL.language) AS LanguageCount
FROM (
  SELECT m.content_id, m.movie_title AS Title,
    'Movie' AS ContentType FROM movie m
  UNION ALL
  SELECT e.content_id, e.series_title AS Title,
    'Episode' AS ContentType FROM episode e
) AS T
  JOIN content_language CL ON T.content_id = CL.content_id
GROUP BY T.Title, T.ContentType
ORDER BY LanguageCount DESC, T.Title;
```

	Title	ContentType	LanguageCount
►	Andor	Episode	5
	The Mandalorian	Episode	5
	Ahsoka	Episode	4



LESSONS LEARNT

Importance of naming conventions for entities.

Proper normalization ensures each entity is focused on one concept, simplifying queries.

Translating the ER model into LD diagram deepened understanding of normalization and key constraints.

Standard relationship names help with SQL readability and foster team alignment when developing joins and constraints.

The project provided valuable experience in group database development.



THANK YOU